ford..."). Next, to build our error-detection dataset, we evaluate the correctness of each generated response $\hat{y}_i$ by comparing it to the ground-truth label y_i. This comparison yields a correctness label $z_i \in \{0, 1\}$ (1 correct, 0 wrong). The comparison can be done either via automatic heuristics or with the assistance of an instruct-LLM.[2] Our error detection dataset is: $\{(q_i, \hat{y}_i, z_i)\}_{i=1}^{N}$. Note that this dataset is defined based on the analyzed LLM and its generated answers. Any instances where the LLM refuses to answer are excluded, as these can easily be classified as incorrect.

3.2 Experimental Setup

Datasets and models. We perform all experiments on four LLMs: Mistral-7b (Jiang et al., 2023), Mistral-7b-instruct-v0.2 (denoted Mistral-7b-instruct), Llama3-8b (Touvron et al., 2023), and Llama3-8b-instruct. We consider 10 different datasets spanning various domains and tasks: TriviaQA (Joshi et al., 2017), HotpotQA with/without context (Yang et al., 2018), Natural Questions (Kwiatkowski et al., 2019), Winobias (Zhao et al., 2018), Winogrande (Sakaguchi et al., 2021), MNLI (Williams et al., 2018), Math (Sun et al., 2024), IMDB review sentiment analysis (Maas et al., 2011), and a dataset of movie roles (movies) that we curate. We allow unrestricted response generation to mimic real-world LLM usage, with answers decoded greedily. For more details on the datasets and the prompts used to generate answers, refer to Appendix A.3.

Performance metric. We measure the area under the ROC curve to evaluate error detectors, providing a single metric that reflects their ability to distinguish between positive and negative cases across many thresholds, balancing sensitivity (true positive rate) and specificity (false positive rate).

Error detection methods. We compare methods from both uncertainty and hallucinations literature.

- **Aggregated probabilities / logits:** Previous studies (Guerreiro et al., 2023; Kadavath et al., 2022; Varshney et al., 2023; Huang et al., 2023b) aggregate output token probabilities or logits to score LLM confidence for error detection. We implement several methods from the literature, calculating the minimum, maximum, or mean of these values. The main paper reports results for the most common approach, **Logits-mean**, and the best-performing one, **Logits-min**, with additional baselines in Appendix B.
- **P(True):** Kadavath et al. (2022) showed that LLMs are relatively calibrated when asked to evaluate the correctness of their generation via prompting. We implement this evaluation using the same prompt.
- **Probing:** Probing classifiers involve training a small classifier on a model's intermediate activations to predict features of processed text (Belinkov, 2021). Recent studies show their effectiveness for error detection in generated text (Kadavath et al., 2022, *inter alia*). An intermediate activation is a vector $h_{l,t}$ from a specific LLM layer l and (either read or generated) token t. Thus, each LLM generation produces multiple such activations. Following prior work, we use a linear probing classifier for error detection (Li et al., 2024, inter alia) on static tokens: the last generated token ($h_{l,-1}$), the one before it ($h_{l,-2}$), and the final prompt token ($h_{l,k}$). The layer l is selected per token based on validation set performance.

For further details on the implementation of each method, refer to Appendix A.4.

Exact Answer Tokens. Existing methods often overlook a critical nuance: the token selection for error detection, typically focusing on the last generated token or taking a mean. However, since LLMs typically generate long-form responses, this practice may miss crucial details (Brunner et al., 2020). Other approaches use the last token of the prompt (Slobodkin et al., 2023, *inter alia*), but this is inherently inaccurate due to LLMs' unidirectional nature, failing to account for the generated response and missing cases where different sampled answers from the same model vary in correctness. We investigate a previously unexamined token location: the *exact answer tokens*, which represent the most meaningful parts of the generated response. We define exact answer tokens as those whose modification alters the answer's correctness, disregarding subsequent generated content.[3] Figure 1 illustrates the different token locations. In the following experiments, we implement

[2]For most datasets, we use heuristics to predict correctness, except for one case. See Appendix A.2.

[3]In practice, we do not use this definition for extracting the exact answer, but rather an instruct model in a few-shot setting. Still, the definition is useful to manually verify that automatic extractions work as expected.

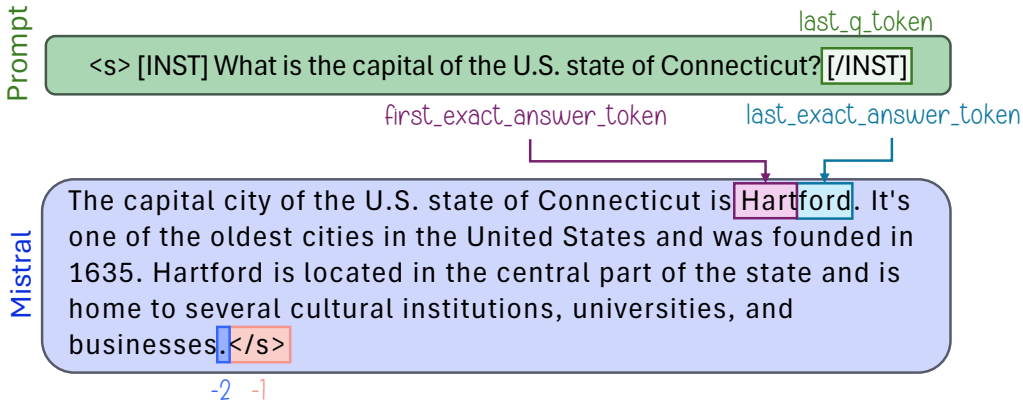
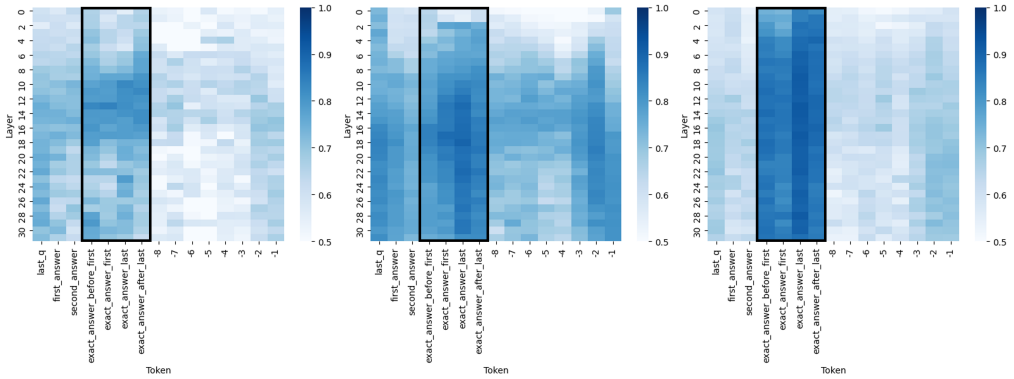


Figure 1: Example for the input and LLM output from the TriviaQA dataset, and the names of the tokens that can be probed.



(a) TriviaQA (b) Winobias (c) Math

Figure 2: AUC values of a probe error detector across layers and tokens, Mistral-7b-instruct. Generation proceeds from left to right, with detection performance peaking at the exact answer tokens.

each error detection method with an "exact answer" version, demonstrating that it often improves performance, especially in probing. Implementation details for detecting the exact answer token are given in Appendix A.2.

3.3 Results

Patterns of truthfulness encoding. We first focus on probing classifiers to gain insights into the internal representations of LLMs. Specifically, we analyze the effects of layer and token selection on the error detection performance of these probing classifiers. By systematically probing all model layers, starting from the last question token to the final generated token, we observe consistent truthfulness encoding patterns. Figure 2 shows AUC metrics of probes across Mistral-7b-Instruct layers and tokens. Middle to later layers often yield the most effective probing results (see Appendix B for more datasets and models), aligning with previous studies on truthfulness encoding (Burns et al., 2022; CH-Wang et al., 2023) and transformer representations (nostalgebraist, 2020; Meng et al., 2022; Geva et al., 2023). Regarding tokens, a strong truthfulness signal appears immediately after the prompt, suggesting that this representation encodes information on the model's general ability to answer the question correctly. This signal weakens as text generation progresses but peaks again at the exact answer tokens. Towards the end of the generation process, signal strength rises again, though it remains weaker than at the exact answer tokens. These patterns are consistent across nearly all datasets and models (see Appendix B), suggesting a general mechanism by which LLMs encode and process truthfulness during text generation.

Table 1: Comparison of error detection techniques using AUC metric, across different models and datasets. The best-performing method is **bolded**. Using exact answer tokens is useful for many cases, especially probing.

	Mistral-7b-Instruct			**Llama 3-8b-Instruct**		
	TriviaQA	Winobias	Math	TriviaQA	Winobias	Math
Logits-mean	$0.60_{\pm 0.009}$	$0.56_{\pm 0.017}$	$0.55_{\pm 0.029}$	$0.66_{\pm 0.005}$	$0.60_{\pm 0.026}$	$0.75_{\pm 0.018}$
Logits-mean-exact	$0.68_{\pm 0.007}$	$0.54_{\pm 0.012}$	$0.51_{\pm 0.005}$	$0.71_{\pm 0.006}$	$0.55_{\pm 0.019}$	$0.80_{\pm 0.021}$
Logits-min	$0.63_{\pm 0.008}$	$0.59_{\pm 0.012}$	$0.51_{\pm 0.017}$	$0.74_{\pm 0.007}$	$0.61_{\pm 0.024}$	$0.75_{\pm 0.016}$
Logits-min-exact	$0.75_{\pm 0.006}$	$0.53_{\pm 0.013}$	$0.71_{\pm 0.009}$	$0.79_{\pm 0.006}$	$0.61_{\pm 0.019}$	$0.89_{\pm 0.018}$
p(True)	$0.66_{\pm 0.006}$	$0.45_{\pm 0.021}$	$0.48_{\pm 0.022}$	$0.73_{\pm 0.008}$	$0.59_{\pm 0.020}$	$0.62_{\pm 0.017}$
p(True)-exact	$0.74_{\pm 0.003}$	$0.40_{\pm 0.021}$	$0.60_{\pm 0.025}$	$0.73_{\pm 0.005}$	$0.63_{\pm 0.014}$	$0.59_{\pm 0.018}$
Probe @ token						
Last generated [-1]	$0.71_{\pm 0.006}$	$0.82_{\pm 0.004}$	$0.74_{\pm 0.008}$	$0.81_{\pm 0.005}$	$0.86_{\pm 0.007}$	$0.82_{\pm 0.016}$
Before last generated [-2]	$0.73_{\pm 0.004}$	$0.85_{\pm 0.004}$	$0.74_{\pm 0.007}$	$0.75_{\pm 0.005}$	$0.88_{\pm 0.005}$	$0.79_{\pm 0.020}$
End of question	$0.76_{\pm 0.008}$	$0.82_{\pm 0.011}$	$0.72_{\pm 0.007}$	$0.77_{\pm 0.007}$	$0.80_{\pm 0.018}$	$0.72_{\pm 0.023}$
Exact	$\mathbf{0.85}_{\pm 0.004}$	$\mathbf{0.92}_{\pm 0.005}$	$\mathbf{0.92}_{\pm 0.008}$	$\mathbf{0.83}_{\pm 0.002}$	$\mathbf{0.93}_{\pm 0.004}$	$\mathbf{0.95}_{\pm 0.027}$

Error Detection Results. Next, we evaluate various error detection methods by comparing their performance with and without the use of exact answer tokens. Table 1 compares the AUC across three representative datasets (additional datasets and models in Appendix B, showing consistent patterns). Here we present results for the last exact answer token, which outperformed both the first exact answer token and the one preceding it, while the token following the last performed similarly. Incorporating the exact answer token improves the different error detection methods in almost all datasets. Notably, our probing technique (bottom line) consistently outperforms all other baselines across the board. While we did not compare all existing error detection methods, the primary conclusion is that information about truthfulness is highly localized in specific generated tokens, and that focusing on exact answer tokens leads to significant improvements in error detection.

4 Generalization Between Tasks

The effectiveness of a probing classifier in detecting errors suggests that LLMs encode information about the truthfulness of their outputs. This supports using probing classifiers for error detection in production, but their generalizability across tasks remains unclear. While some studies argue for a universal mechanism of truthfulness encoding in LLMs (Marks & Tegmark, 2023; Slobodkin et al., 2023), results on probe generalization across datasets are mixed (Kadavath et al., 2022; Marks & Tegmark, 2023; CH-Wang et al., 2023; Slobodkin et al., 2023; Levinstein & Herrmann, 2024)–observing a decline in performance, yet it remains significantly above random chance. Understanding this is essential for real-world applications, where the error detector may encounter examples that significantly differ from those it was trained on. Therefore, we explore whether a probe trained on one dataset can detect errors in others.

Our generalization experiments are conducted between all of the ten datasets discussed in Section 3, covering a broader range of reaslistic task settings than previous work. This breadth of experiments has not been previously explored, and is crucial considering the mixed findings in previous work. We select the optimal token and layer combination for each dataset, train all probes using this combination on other datasets, and then test them on the original dataset. We evaluate generalization performance using the absolute AUC score, defined as $\max(\text{auc}, 1 - \text{auc})$, to also account for cases where the learned signal in one dataset is reversed in another.

Results. Figure 3a shows the generalization results for Mistral-7b-instruct, with similar patterns observed for other LLMs in Appendix C. In this context, values above 0.5 indicate successful generalization. At first glance, the results appear consistent with previous research: most heatmap values exceed 0.5, implying some degree of generalization across tasks. This observation supports the existence of a universal mechanism for decoding truthfulness, since the same linear directions—captured by the probe—encode truthfulness information across many datasets. However, upon closer inspection, it turns out that most of this performance can be achieved by logit-based truthfulness detection,